



Torque and Angular Momentum

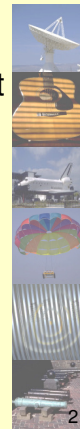
Chapter 11

Torque (cross-product)
Angular Momentum

Do all the Sample Problems in the relevant sections

Cross Products

- Review the rules regarding cross-products
- Review application of the right hand rule to determine the direction of the resultant vector of the cross-product of two vectors.
- Chapter 3 in HRW



Torque

- Torque is defined as: $\tau = (r)(F \sin \phi) = (r \sin \phi)(F)$
- Torque can be written as a vector cross product:
$$\vec{\tau} = \vec{r} \times \vec{F}$$
- Right hand rule** - move $\vec{F}$ to O , curl fingers from $\vec{r}$ to $\vec{F}$, thumb shows direction of $\vec{\tau}$.
- Read section 11-6 for more information and Sample Problems



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Problem 25

- Force $\vec{F} = (2.0 \text{ N})\hat{i} - (3.0 \text{ N})\hat{k}$ acts on a pebble with position vector $\vec{r} = (0.5 \text{ m})\hat{j} - (2.0 \text{ m})\hat{k}$ relative to the origin. In unit-vector notation, what is the resulting torque on the pebble about (a) the origin and (b) the point (2.0m, 0, -3.0 m)?

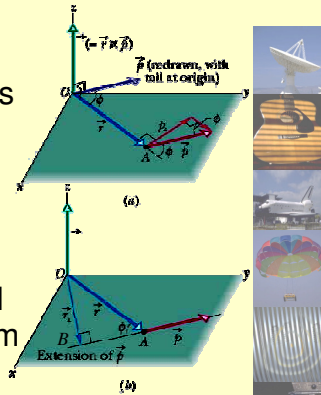


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Angular Momentum

- Conservation of linear momentum is a powerful tool
- Particle of mass m with $\vec{p} = m\vec{v}$ passes through point A in the xy plane.
- The angular momentum of this particle w.r.t. the origin O is defined as

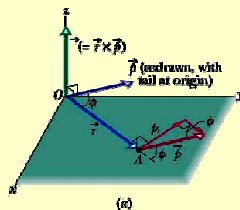
$$\vec{\ell} = \vec{r} \times \vec{p} = m(\vec{r} \times \vec{v})$$
- $\vec{r}$ is the position vector w.r.t. the origin O
- Units: $\text{kg} \cdot \text{m}^2/\text{s}$
- Particle doesn't have to rotate around O in order to have angular momentum about O .



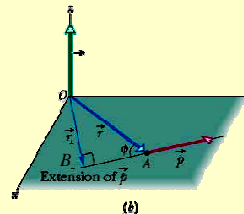
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Angular Momentum

- Direction – right-hand rule.
- Magnitude: $\ell = rmv \sin \phi$



$$\ell = rp_{\perp} = rmv_{\perp}$$



$$\ell = r_{\perp} p = r_{\perp} mv$$

- Read section 11-7 for more information and Sample Problems



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Problem 29

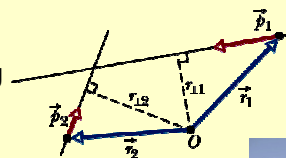
- At a certain time, force $\vec{F} = 4.0\hat{j}$ N acts on a 0.25 kg object that has a position vector $\vec{r} = (2.0\hat{i} - 2.0\hat{k})$ m and velocity vector $\vec{v} = (-5.0\hat{i} + 5.0\hat{k})$ m/s. About the origin and in unit vector notation, what are (a) the object's angular momentum and (b) the torque acting on the object?



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Sample Problem 11-4

- The figure shows an overhead view of two particles moving at constant momentum along horizontal paths. Particle 1, with momentum magnitude $p_1 = 5.0$ kg·m/s has position vector $\vec{r}_1$ and will pass 2.0 m from point O . Particle 2, with momentum magnitude $p_2 = 2.0$ kg·m/s has position vector $\vec{r}_2$ and will pass 4.0 m from point O . What are the magnitude and direction of the net angular momentum $\vec{L}$ about point O of the two particle system?



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Newton's 2nd Law in Angular Form

- Linear - single particle $\sum \vec{F}_{net} = \frac{d\vec{p}}{dt}$
- Angular - single particle $\sum \vec{\tau}_{net} = \frac{d\vec{\ell}}{dt}$
- The vector sum of all the torques on a particle is equal to the rate of change of the angular momentum of the particle.
- Read section 11-7 for more information and Sample Problems



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Angular momentum of a system of particles

- $\vec{L} = \vec{\ell}_1 + \vec{\ell}_2 + \vec{\ell}_3 + \dots \vec{\ell}_n = \sum_{i=1}^n \vec{\ell}_i$
 $\frac{d\vec{L}}{dt} = \sum_{i=1}^n \vec{\tau}_{net,i}$
- System of particles $\vec{\tau}_{net} = \frac{d\vec{L}}{dt}$
- For rigid bodies $L = I\omega$
- Read section 11-9/10 for more information and Sample Problems



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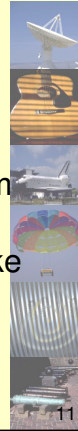
Conservation of Angular Momentum

- If the $\vec{\tau}_{net} = 0$, (isolated system)
- $\vec{L} = \text{constant}$
- No matter what change takes place within the system
- **Law of conservation of angular momentum:**

$$\vec{L}_i = \vec{L}_f$$

$$I_i \omega_i = I_f \omega_f$$

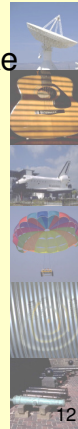
- If **any component of the net external torque** on a system is **zero**, the angular momentum $\vec{L}$ of the system along that axis remains constant, no matter what changes take place within the system.
- Read section 11-9/10 for more information and Sample Problems



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Sample Problem 11-8

- A cockroach with mass m rides on a disk of mass $6.00m$ and radius R . The disk rotates like a merry-go-round around its central axis at angular speed $\omega = 1.50 \text{ rad/s}$. the cockroach is initially at radius $0.800R$, but then crawls out to the rim of the disk. Treat the cockroach like a particle. What then is the angular speed?



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